

KESWICK, United Kingdom

WMO: 032120

Lat: 54.6139N

Lon: 3.1567W

Elev: 266

StdP: 14.56

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.9	26.8	21.0	15.9	30.5	23.3	17.8	31.6	25.8	45.4	23.6	46.0	1.2	120	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	75.8	64.4	72.0	62.2	69.0	60.7	65.6	73.2	63.6	69.9	61.9	67.3	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	86.0	69.1	61.0	80.8	66.6	59.4	76.4	64.7	30.5	73.3	29.1	69.9	27.8	67.3	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	17.9	15.5	19.4	80.7	3.9	3.7	16.6	83.3	14.4	85.5	12.2	87.5	9.3	90.2		
(5)			19.4	69.5	4.0	3.3	16.6	71.9	14.3	73.9	12.1	75.8	9.2	78.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	40.6	40.7	42.7	46.8	51.9	56.6	59.8	59.0	55.5	50.4	44.9	40.9	
	DBStd	8.73	5.66	5.44	5.18	4.96	5.20	4.61	4.13	3.81	4.26	5.34	5.88	6.82	
	HDD50	1467	293	262	231	120	37	3	0	0	5	60	170	286	
	HDD65	5793	757	680	691	547	408	256	172	191	287	452	604	748	
	CDD50	1170	1	2	5	23	95	200	303	280	169	73	15	4	
	CDD65	21	0	0	0	0	1	4	10	5	1	0	0	0	
	CDH74	183	0	0	0	0	8	39	93	41	2	0	0	0	
	CDH80	19	0	0	0	0	0	5	9	5	0	0	0	0	
Wind	WSAvg	5.7	7.9	7.6	7.4	6.2	5.4	4.3	3.5	3.6	4.0	5.2	6.2	7.1	
Precipitation	PrecAvg	55.7	5.9	4.4	3.9	3.4	3.3	3.8	4.3	4.6	6.1	6.1	6.5		
	PrecMax	70.5	9.3	10.8	6.8	5.5	6.4	8.9	7.8	8.8	6.9	10.9	15.7	18.4	
	PrecMin	39.8	0.4	1.1	1.2	1.4	0.8	1.2	2.0	0.7	0.9	1.7	3.2	1.1	
	PrecStd	8.3	2.3	2.6	1.4	1.1	1.5	1.7	1.3	1.9	1.7	2.8	2.8	3.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	55.8	59.8	69.0	74.7	78.8	81.0	79.7	72.1	64.9	59.0	56.3	
		MCWB	50.9	50.4	50.8	55.9	61.7	66.8	67.5	65.8	63.1	57.9	55.3	52.8	
	2%	DB	52.1	52.5	54.8	62.1	69.9	73.1	76.5	72.8	68.0	61.8	56.8	54.0	
		MCWB	49.0	48.8	49.5	52.6	59.4	63.0	64.8	63.5	60.5	57.1	53.5	51.0	
	5%	DB	50.4	50.7	52.6	58.1	65.7	69.2	72.4	69.0	65.2	60.1	54.8	52.3	
		MCWB	47.2	47.4	48.2	50.5	56.7	60.7	62.7	61.4	58.8	56.0	51.8	49.4	
	10%	DB	48.8	48.9	50.7	55.1	61.6	65.9	68.5	66.2	63.0	58.4	53.1	50.5	
		MCWB	46.0	45.6	46.8	49.1	53.9	58.2	61.1	59.8	57.5	54.6	50.3	47.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	51.5	52.9	56.8	63.5	67.6	69.5	67.1	64.6	60.1	56.4	53.6
			MCDB	53.8	54.1	57.4	65.1	72.5	75.8	79.0	75.7	69.9	62.5	58.4	55.8
2%		WB	49.6	49.5	50.4	53.8	60.0	64.1	66.0	64.5	62.0	58.3	54.0	51.6	
		MCDB	51.7	52.0	53.8	60.3	68.0	71.3	73.9	70.8	66.4	61.0	56.2	53.7	
5%		WB	47.9	47.7	48.7	51.8	57.4	61.3	63.8	62.7	60.0	56.6	52.3	49.8	
		MCDB	50.0	50.4	51.8	56.8	64.2	67.8	70.2	67.3	63.7	59.7	54.5	52.1	
10%		WB	46.1	46.0	47.1	49.9	55.0	59.0	61.8	61.1	58.3	54.9	50.6	47.8	
		MCDB	48.6	48.6	50.3	54.4	60.5	64.5	67.5	65.1	62.1	58.0	52.9	50.4	
Mean Daily Temperature Range		5% DB	MDBR	8.5	9.7	11.4	13.4	15.0	14.3	14.2	13.3	13.1	11.2	9.5	9.6
			MCDBR	8.9	10.2	13.5	19.0	22.0	21.1	22.0	18.8	17.0	11.7	9.2	9.8
	MCWBR		8.6	8.8	9.6	11.9	13.5	13.1	12.5	11.3	10.6	8.9	8.3	9.2	
	5% WB	MCDBR	8.9	9.3	11.7	16.1	20.1	19.1	18.9	16.1	14.1	10.6	9.0	9.8	
		MCWBR	8.8	8.7	9.3	10.9	13.0	12.7	11.7	10.5	9.8	9.0	8.6	9.4	
Clear-Sky Solar Irradiance	taub		0.325	0.336	0.363	0.382	0.386	0.383	0.384	0.371	0.359	0.344	0.320	0.311	
	taud		2.270	2.354	2.308	2.284	2.322	2.356	2.357	2.405	2.444	2.433	2.347	2.249	
	Ebn at Noon		191	237	255	267	272	274	271	268	256	232	194	168	
	Edh at Noon		19	24	31	37	37	36	36	32	27	22	17	16	
All-Sky Solar Radiation	RadAvg		175	387	739	1180	1535	1544	1492	1201	851	456	223	134	
	RadStd		23	43	69	117	118	152	156	89	74	56	23	13	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page